

EXP 4: Build an Artificial Neural Network by implementing the Backpropagation algorithm and test the same using appropriate data sets.

AIM

To build an Artificial Neural Network (ANN) by implementing the Backpropagation algorithm and test it using an appropriate dataset.

SOFTWARE USED

- Google Colab
- Python 3.x
- Pandas Library

PROCEDURE

1. Import the required libraries.
2. Load the dataset.
3. Split the dataset into training and testing sets.
4. Normalize the input features.
5. Build an Artificial Neural Network.
6. Train the network using the Backpropagation algorithm.
7. Test the trained model.
8. Display the accuracy and predictions.

GOOGLE COLAB CODE

```
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

# Load Iris dataset
iris = load_iris()
X = iris.data
y = iris.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# Feature Scaling
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Build ANN
model = Sequential()
model.add(Dense(8, activation='relu', input_shape=(4,)))
model.add(Dense(6, activation='relu'))
model.add(Dense(3, activation='softmax'))

# Compile Model
model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

# Train Model
model.fit(X_train, y_train, epochs=100, verbose=0)

# Test Model
loss, accuracy = model.evaluate(X_test, y_test, verbose=0)
print("Test Accuracy:", accuracy)\

# Prediction
```

```
prediction = model.predict(X_test)
print("\nPredicted Classes:")
print(prediction.argmax(axis=1))
print("\nActual Classes:")
print(y_test)
```

OUTPUT

```
Predicted Classes:
[1 0 2 2 2 0 1 2 1 1 2 0 0 0 0 2 2 1 1 2 0 2 0 2 2 2 2 0 0]

Actual Classes:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]
```

RESULT

Thus, an Artificial Neural Network (ANN) was successfully built using the Backpropagation algorithm, and the model was tested using the Iris dataset. The network achieved high classification accuracy on the test data.